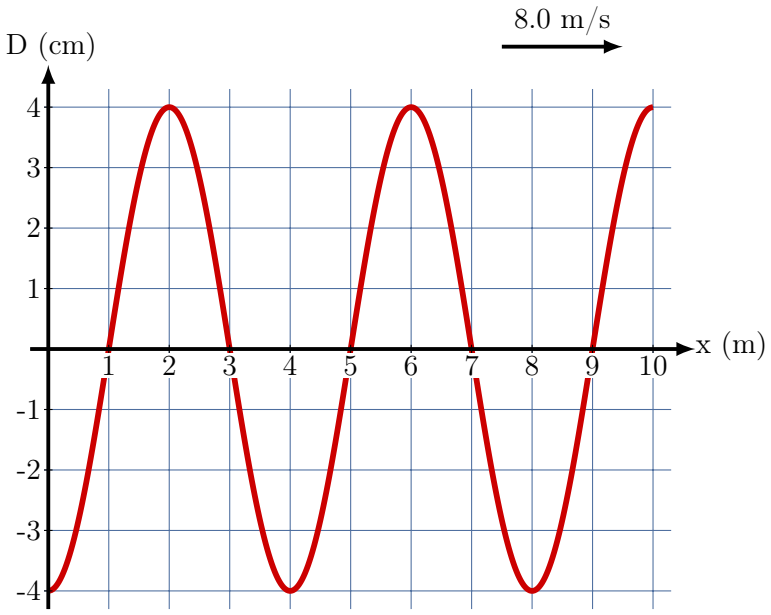
1. A periodic wave travels down a long spring. The wave has a period of $P = 0.02$ s and a wavelength of $\lambda = 5.0$ m.

A. What is the frequency f of the wave? What are the units of f?

B. How fast is the wave traveling?

C. How far does a single crest travel in 0.10 s?

2. The graph below is a **snapshot** of a wave at $t = 0$, $D(x, 0)$. The wave is traveling to the **right** with a speed of 8.0 m/s.



A. Read straight off of the graph:

$A =$ _____ $\lambda =$ _____ $\phi_0 =$ _____

B. Now calculate each of the following. Show your work below.

$k =$ _____ $f =$ _____ $P =$ _____ $\omega =$ _____

C. Write the full wave equation $D(x, t)$ for this wave.

D. Sketch the wave one quarter of a period later, $D(x, \frac{P}{4})$. How far did the wave move?

